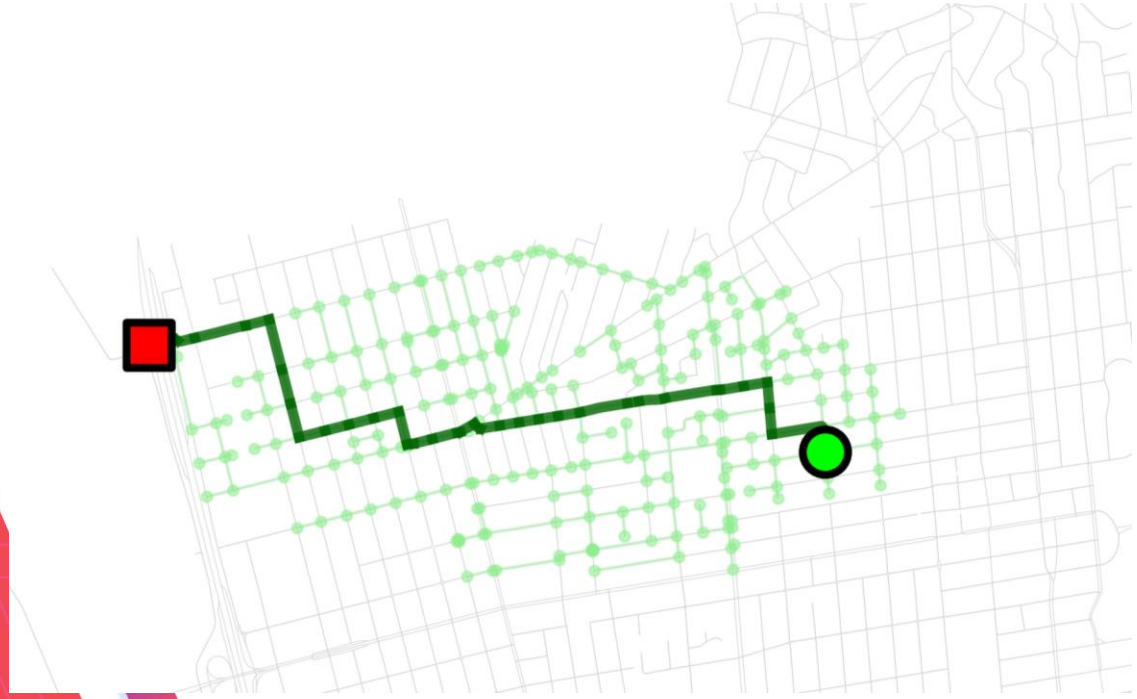
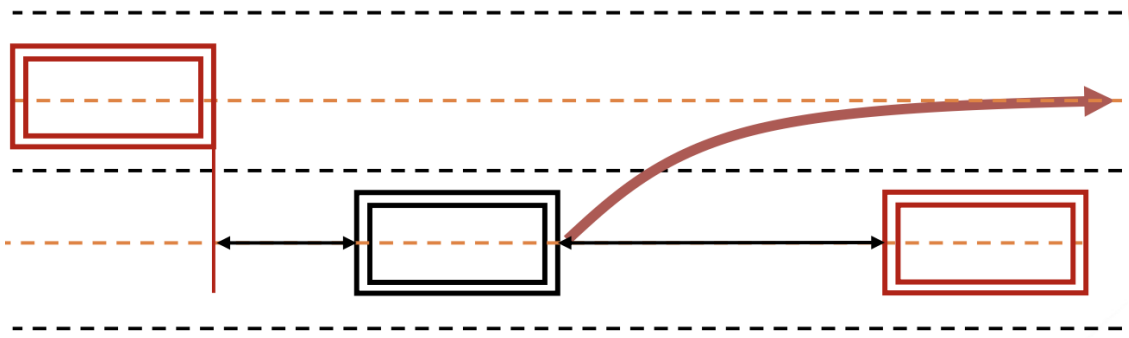
香港城市大學
City University of Hong Kong
Innovating into the Future

CS5493: Tutorial 10
Velocity Planning

Introduction to Planning

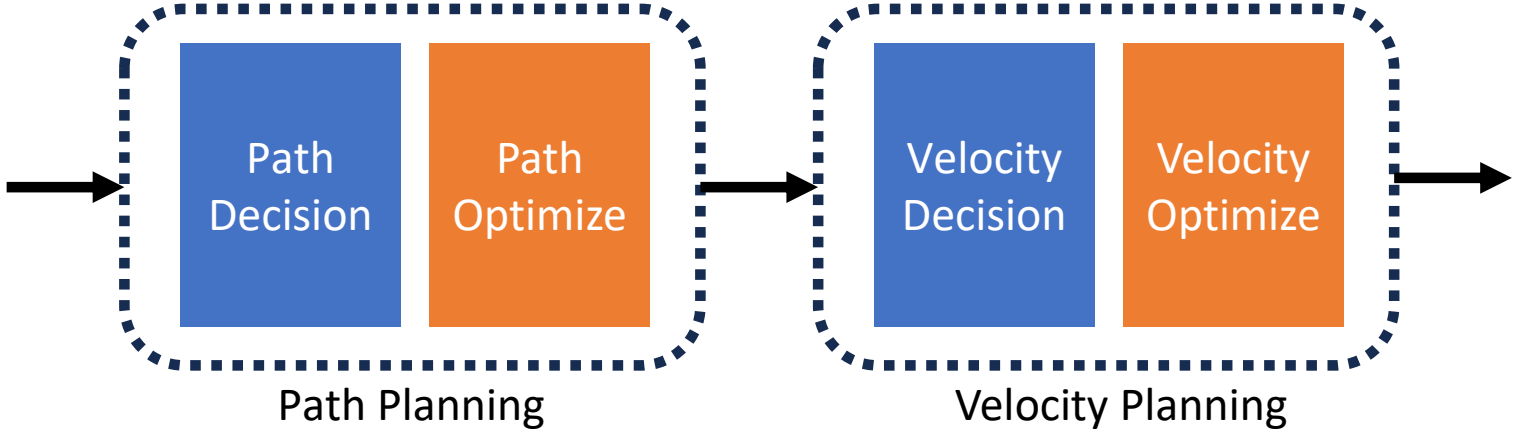
Planning Module focuses on two different scales of path planning.
First is **route navigation**, which generates the route from A to B on a HD Map.
Then, subtle decisions will be made in **trajectory planning**, which focuses on avoiding the obstacles on the planned path, which lane to drive, etc.





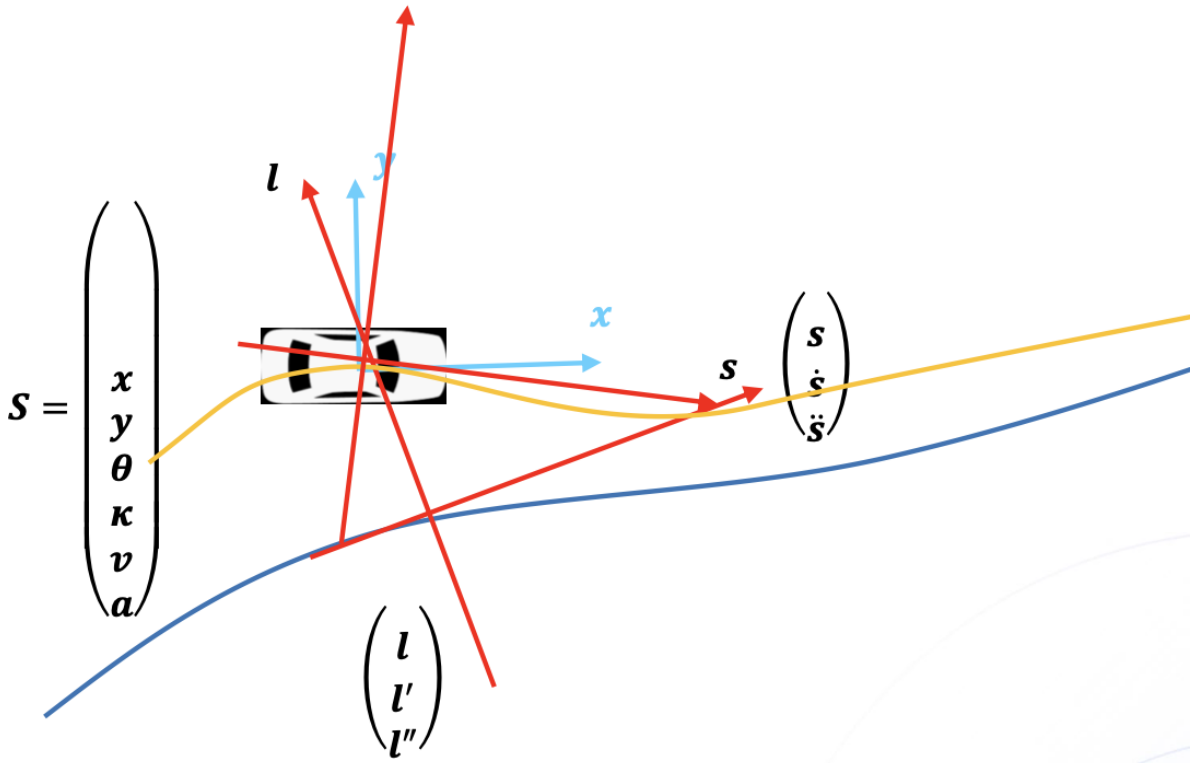
Trajectory Planning Part 2

In trajectory planning, both spatial and temporal trajectory should be considered.
For spatial trajectory planning we call it path planning, which will handle static environment.
For temporal trajectory we define as velocity planning which will handle dynamic obstacles.



Velocity Planning and Optimization

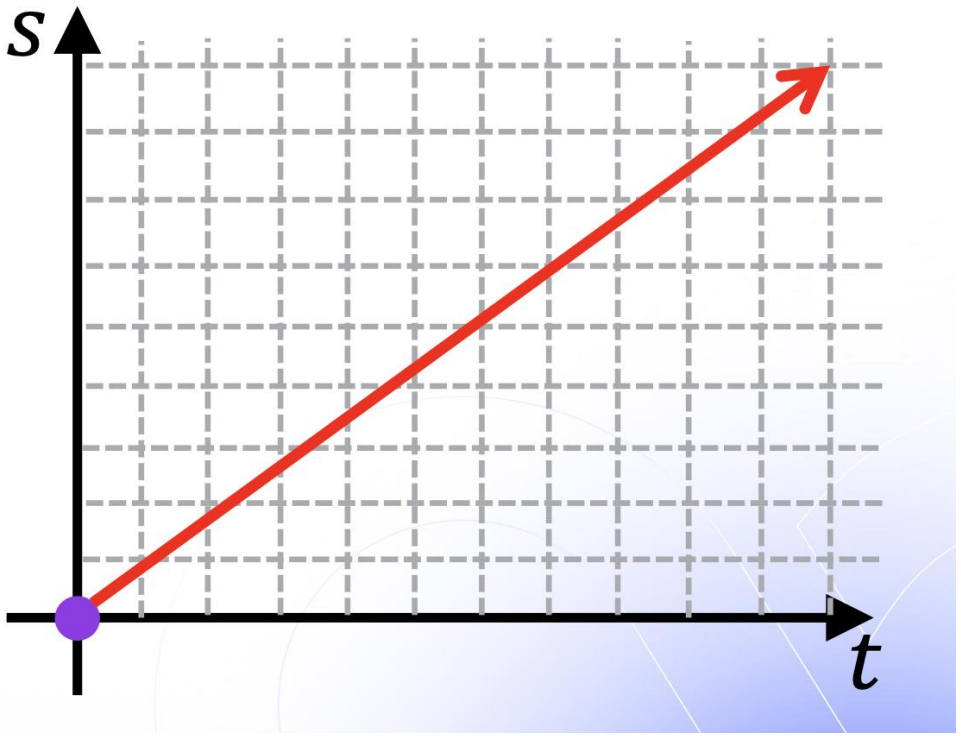
Velocity planning in Apollo is based on Frenet Reference Frame.



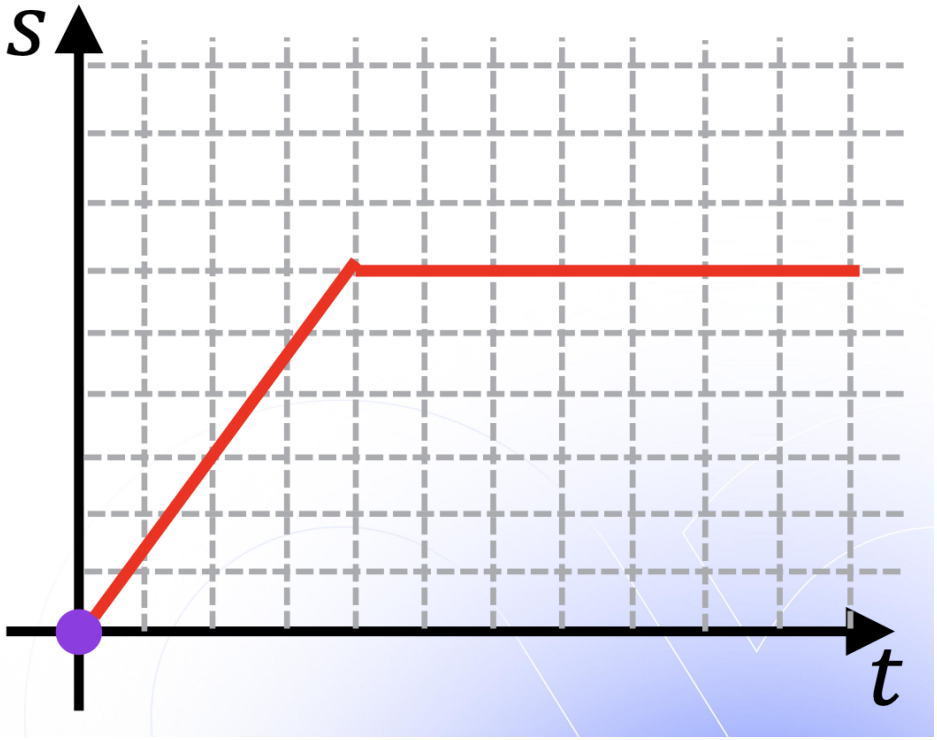
Frenet Reference Frame

Velocity Planning and Optimization

Some classic kinematic states on this reference frame



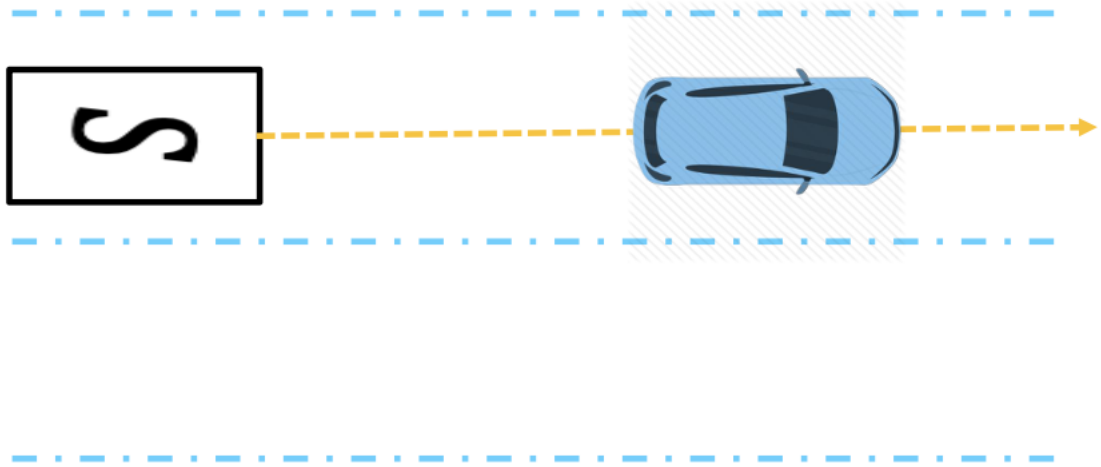
Constant speed along the lane center line



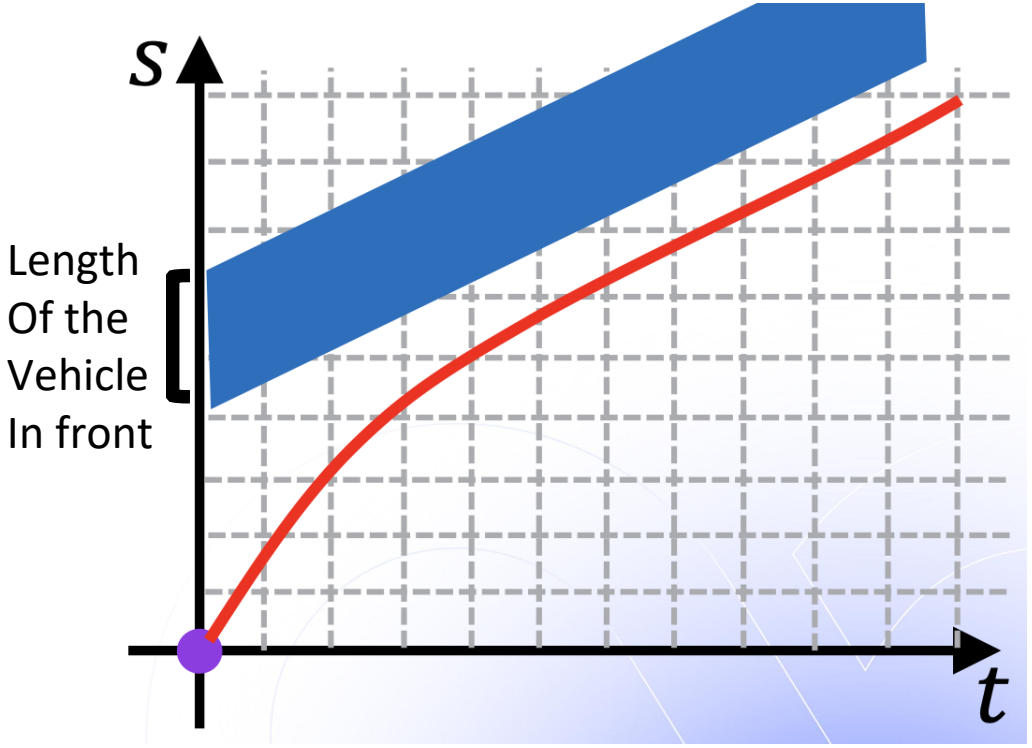
Constant speed along the lane center line
Then suddenly stop

Velocity Planning and Optimization

Some classic kinematic states on this reference frame

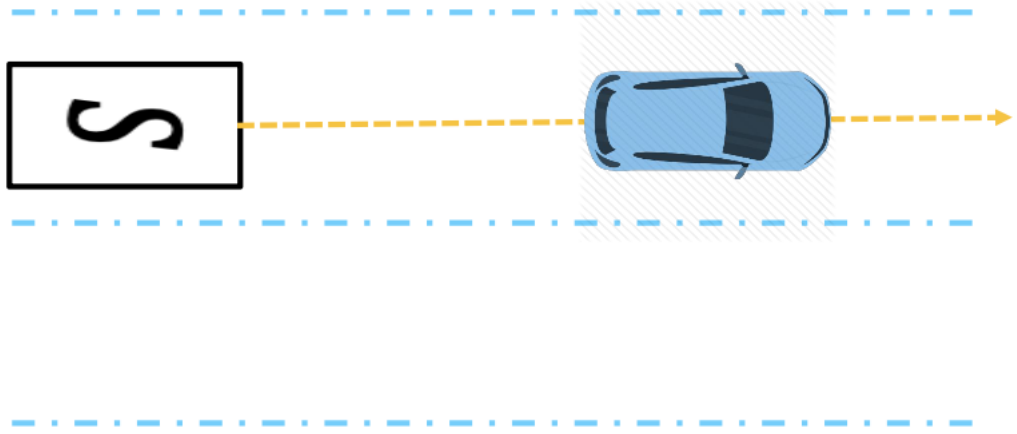


Front vehicle driving with constant speed
And self vehicle follows

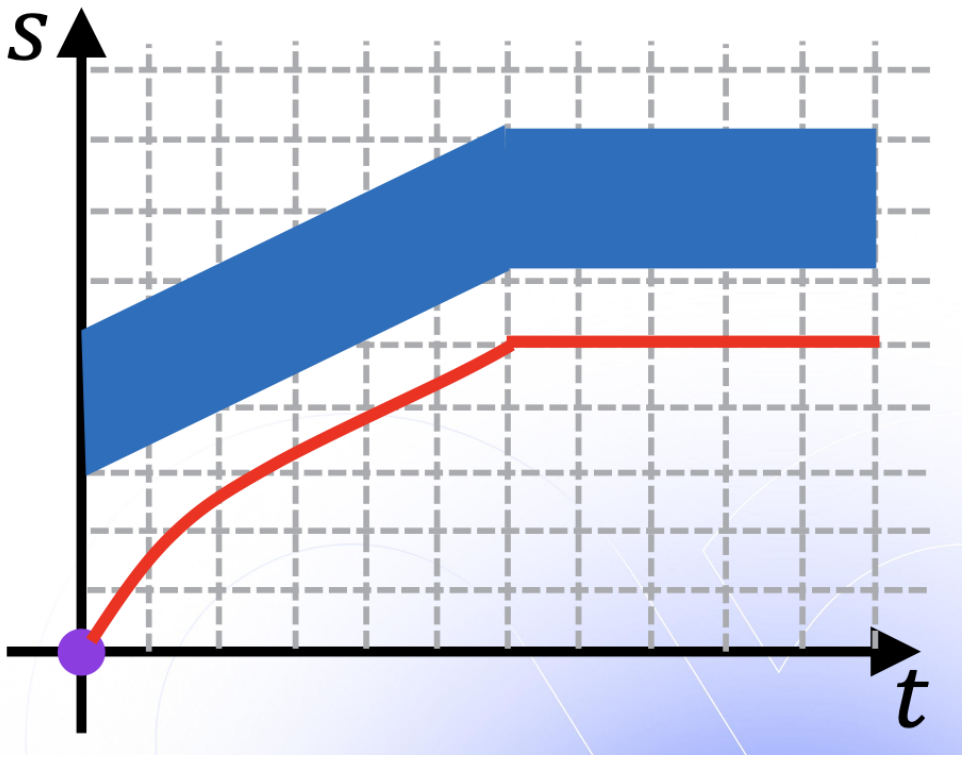


Velocity Planning and Optimization

Some classic kinematic states on this reference frame

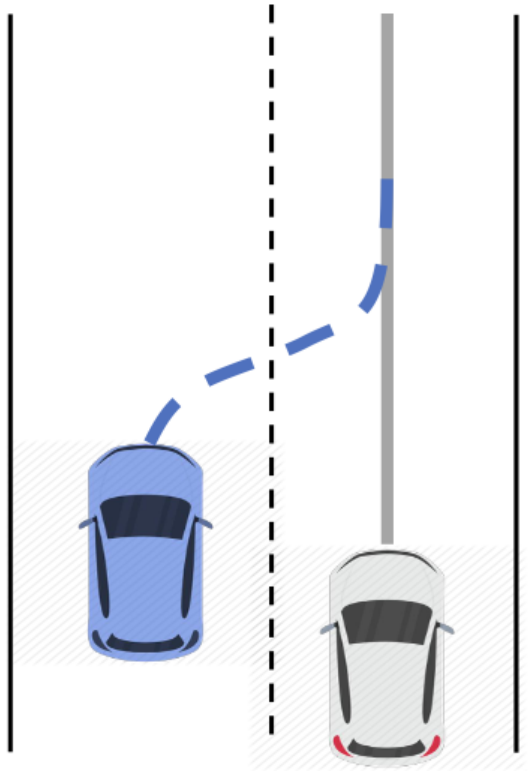


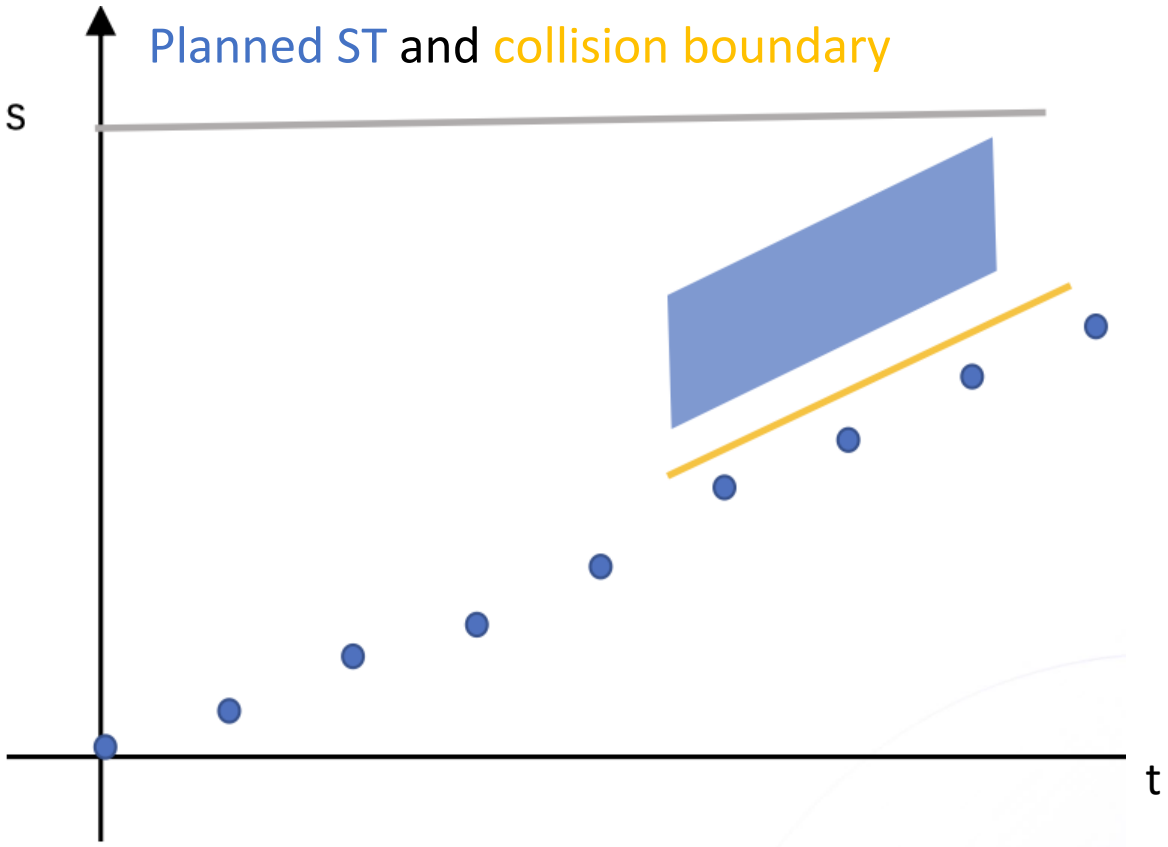
Front vehicle driving with constant speed stops
And self vehicle stops



Velocity Planning and Optimization

Some classic kinematic states on this reference frame



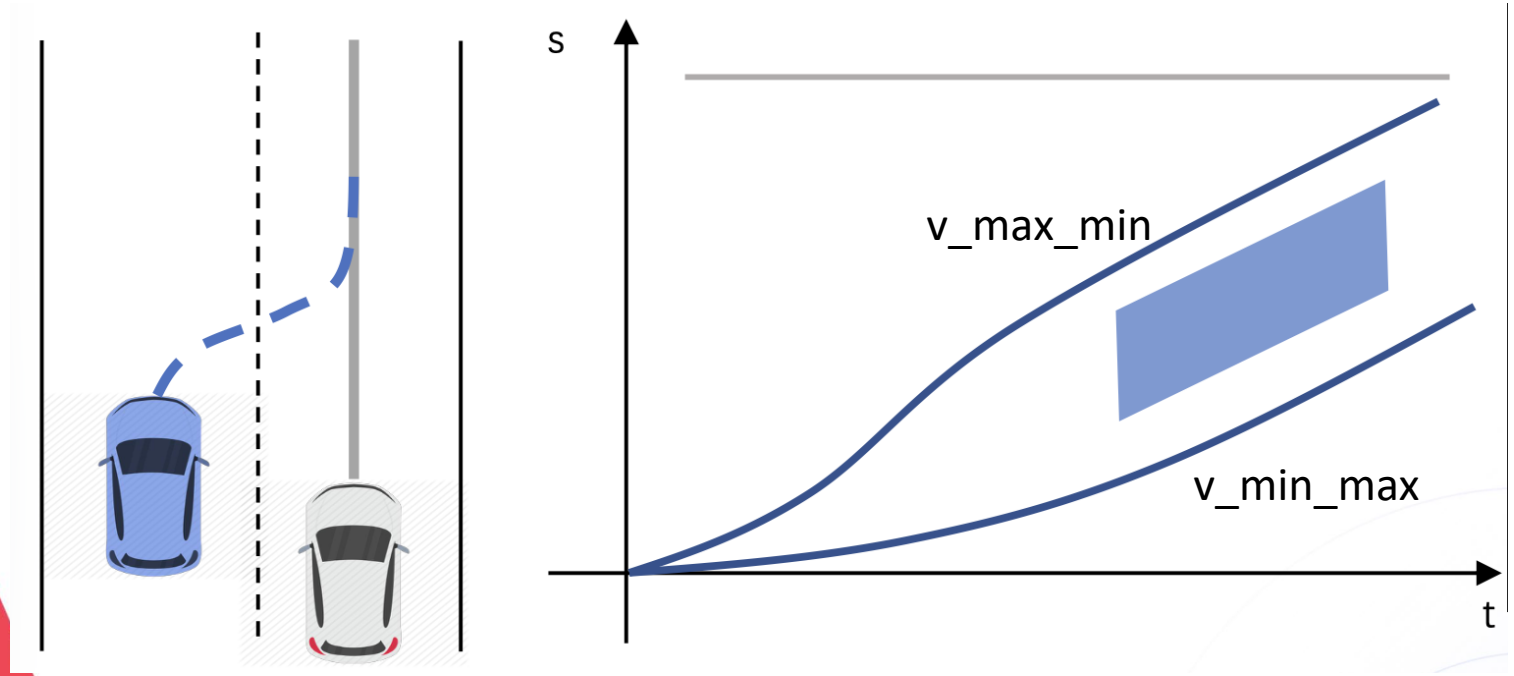


Drivable Area in Velocity Planning

Velocity Planning and Optimization

Velocity planning and optimization workflow

SPEED_BOUNDS_PRIORI_DECIDER

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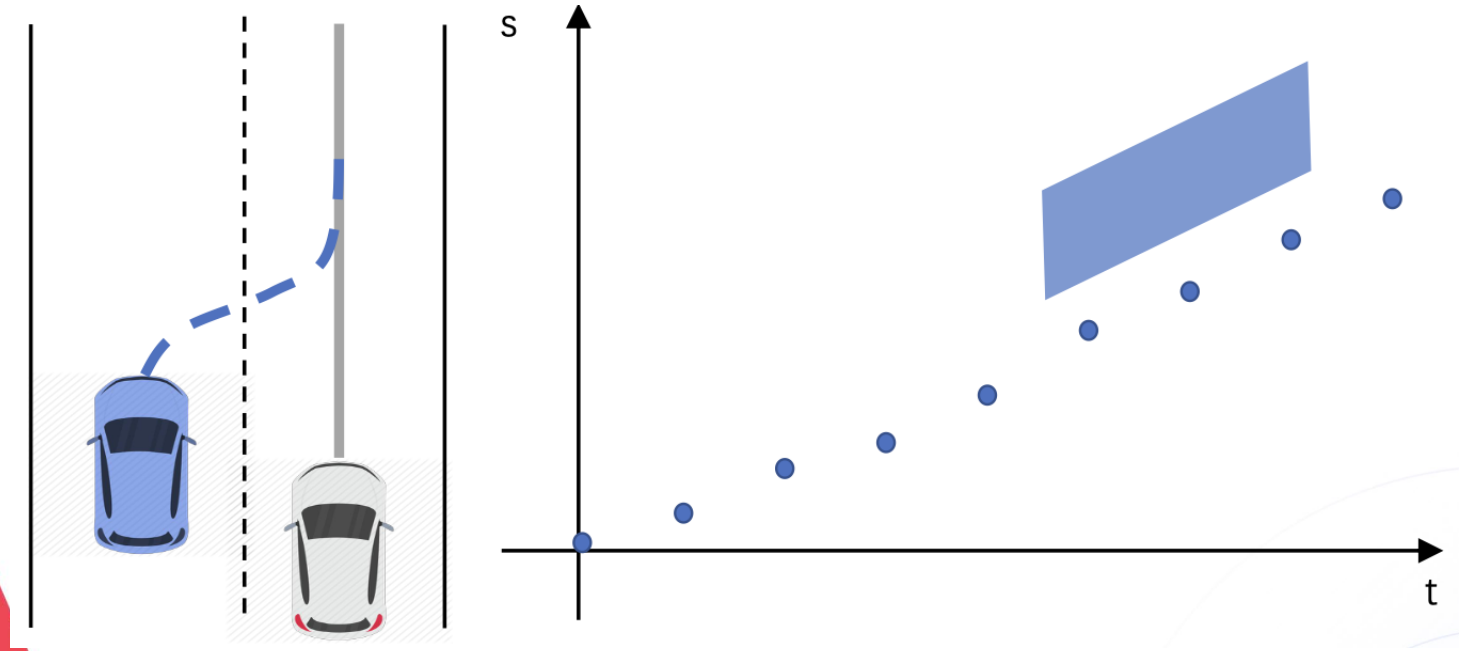
Velocity Planning and Optimization

Velocity planning and optimization workflow

SPEED_HEURISTIC_OPTIMIZER

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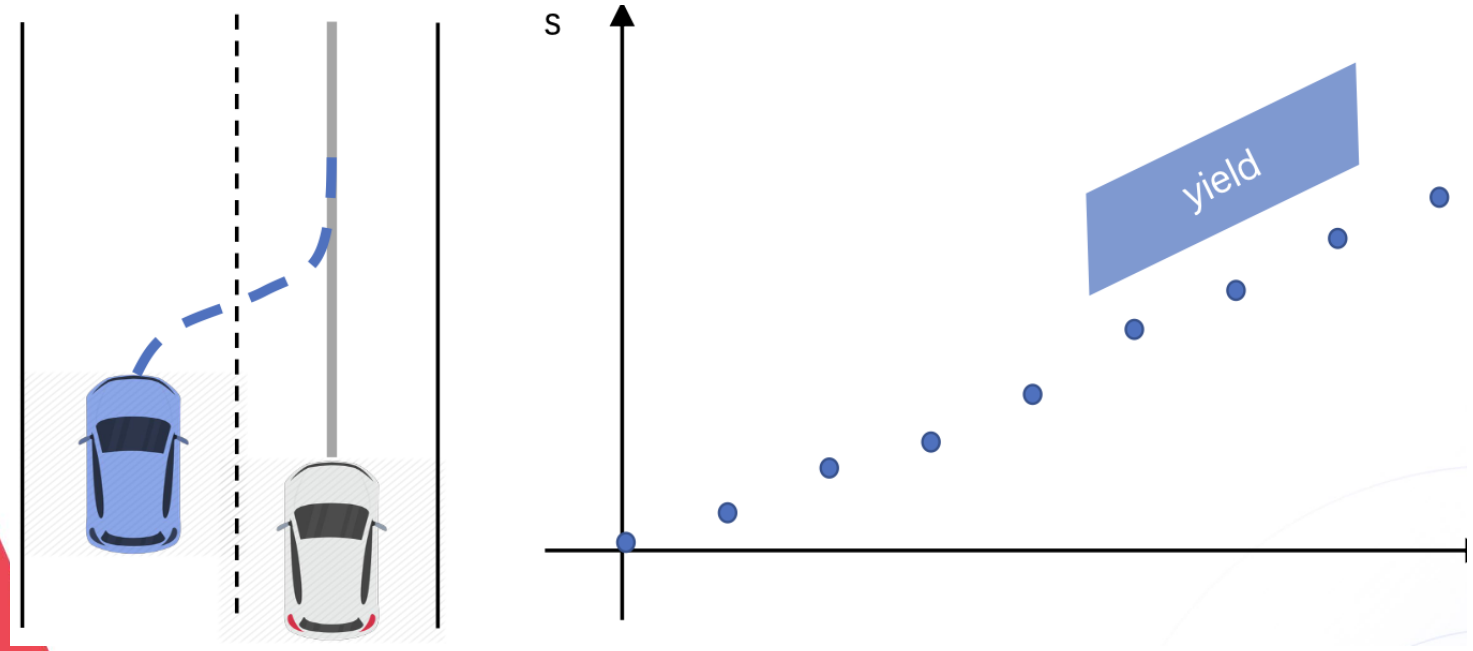
Velocity Planning and Optimization

Velocity planning and optimization workflow

SPEED_DECIDER

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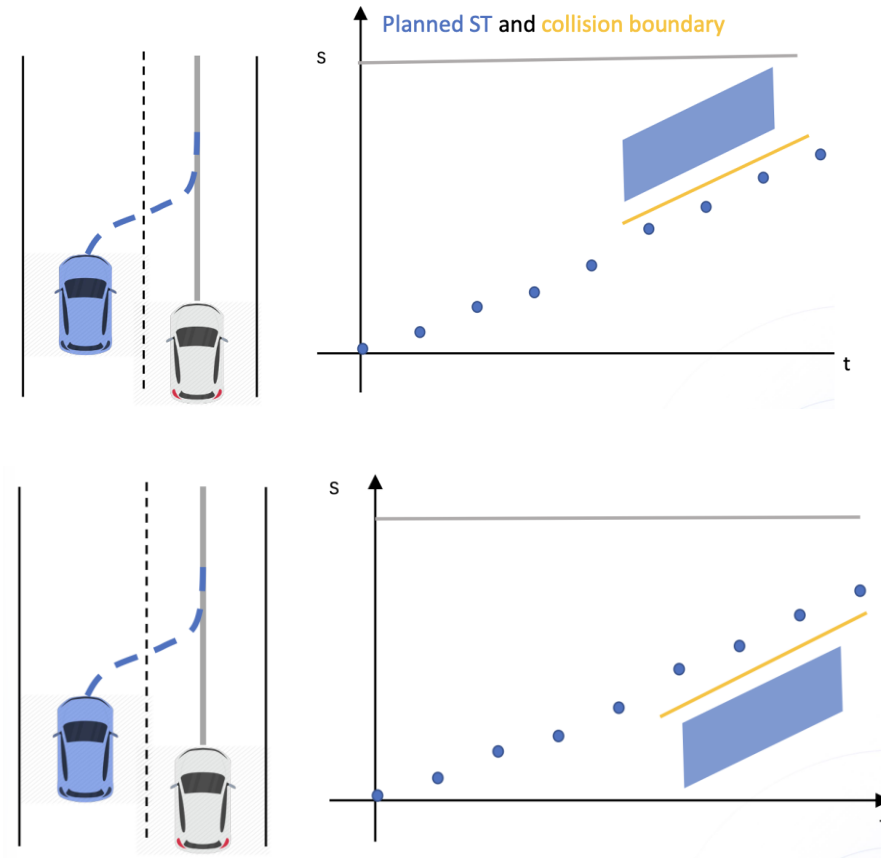
Velocity Planning and Optimization

Velocity planning and optimization workflow

SPEED_BOUNDS_FINAL_DECIDER

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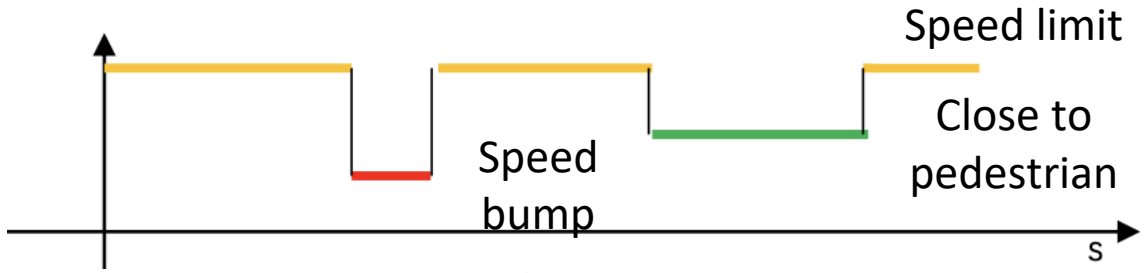
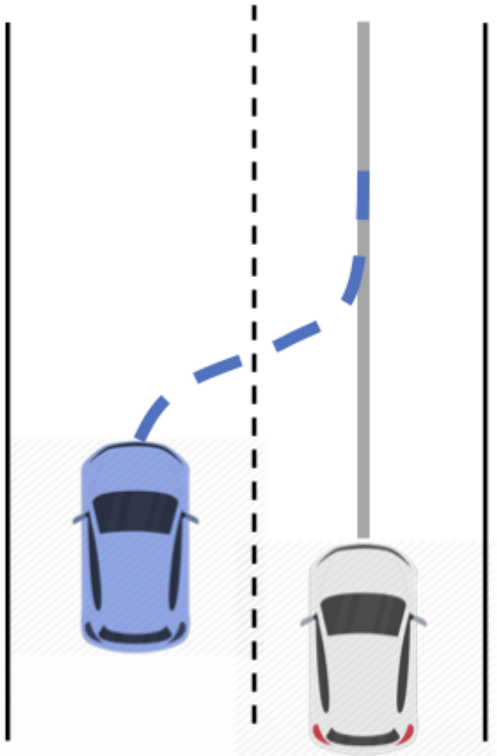
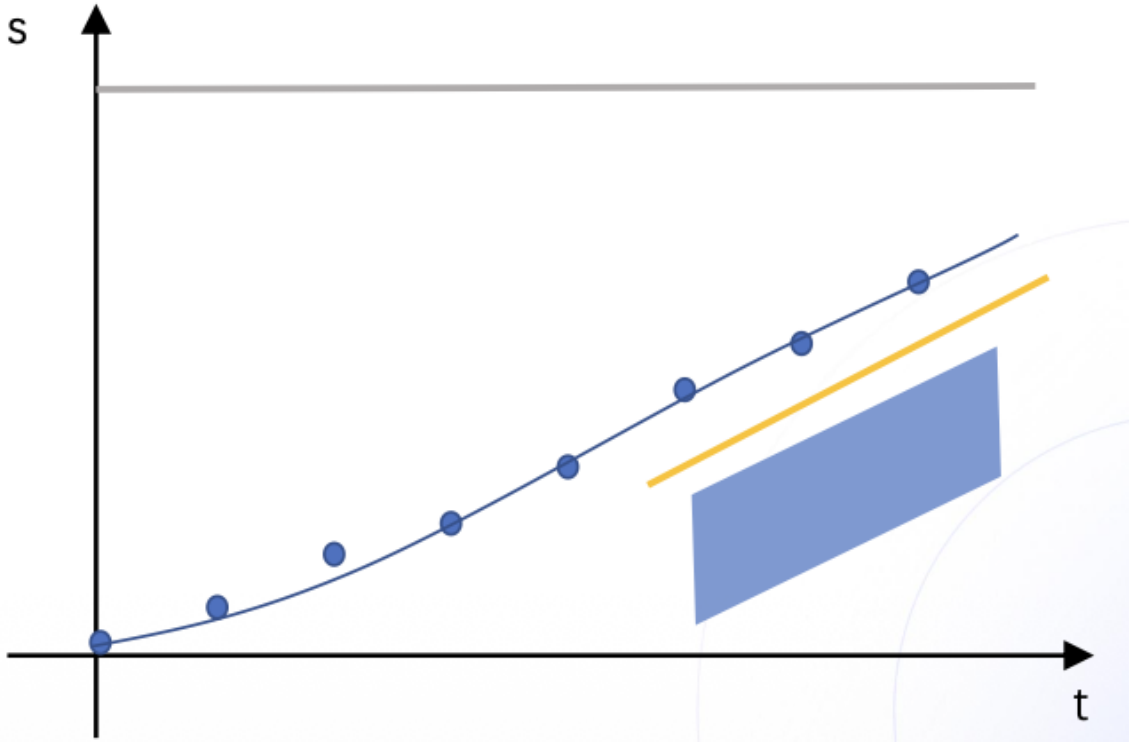
Velocity Planning and Optimization

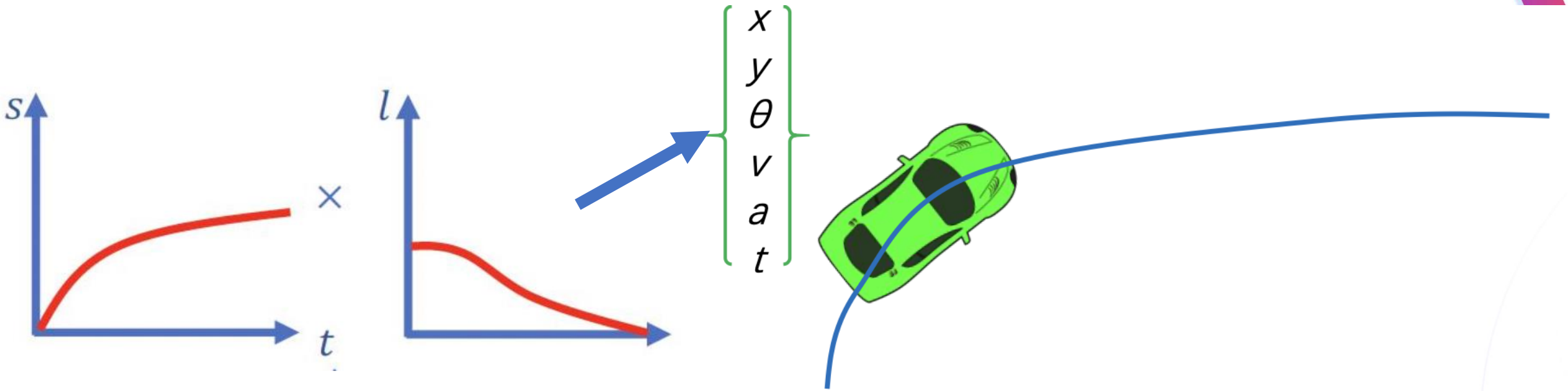
Velocity planning and optimization workflow

PIECEWISE_JERK_NONLINEAR_SPEED_OPTIMIZER







Velocity Planning and Optimization

Finally output the planned trajectory under Euclidean reference frame based on the output of the former workflow.